Cell : The Unit of Life

1. **Given below are two statements :**
   Statement-I : Microbodies are present in both animal and plants cells.
   Statement-II : Eukaryotic cell contain linear ds DNA in chloroplast and mitochondria.
   (1) Both Statement-I and Statement-II are incorrect.
   (2) Statement-I is correct but Statement-II is incorrect.
   (3) Both Statement-I and Statement-II are correct.
   (4) Statement-I is incorrect but Statement II are correct.
2. **Which option is correct regarding eukaryotic Cell:**
   (1) Cytoplasm of Eukaryotic cell contain 70 S and 80S ribosome.
   (2) Eukaryotic cell contain linear ds DNA in chloroplast and mitochondria.
   (3) Only 80S ribosomes are found in cytoplasm, mitochondria and chloroplast of eukaryotic cell.
   (4) 80S ribosomes are present in cytoplasm of eukaryotic cell only.
3. **Hydrolytic enzymes of lysosomes are activated at _______ pH.**
   (1) acidic
   (2) neutral
   (3) basic
   (4) All of the above
4. **Match the following column I and column-II**

Column I	Column II
a. Protease	**i. Lipids**
b. Lipase	**ii. Proteins**
c. Carbohydrase	**iii. Carbohydrates**

   (1) a-ii, b-i, c-iii
   (2) a-i, b-ii, c-iii
   (3) a-ii, b-iii, c-i
   (4) a-iii, b-ii, c-i
5. **Identify for correct set of statements for endoplasmic reticulum**
   (a) ER is known as system of membranes
   (b) It has cis-trans face
   (c) Ribosomes are present on the surface of RER
   (d) SER are the sites of synthesis of lipids and cholesterols
   Choose the correct answer from the given options below.
   (1) all a, b, c & d are correct
   (2) b, c and d are correct
   (3) only c and d are correct
   (4) a, c and d are correct
6. **In __ the __ vacuole is important for excretion.**
   (1) Amoeba, gas
   (2) Bacteria, contractile
   (3) Amoeba, contractile
   (4) Bacteria, gas
7. **Read the following statements.**
   I. It contains water, sap, excretory product and other unwanted materials.
   II. It is bound by a single membrane called tonoplast.
   III. In plant cells, it can occupy up to 90% of cellular volume.
   IV. It content forms cell sap.
   V. It maintains turgor pressure.
   The above features are attributed to:
   (1) Lysosome
   (2) Vacuole
   (3) Peroxisome
   (4) Food

8. **Given below are two statements:**

Statement-I : Mitochondria and chloroplasts are called semi autonomous cell organelles.

Statement-II : Golgi complex is called suicidal bag of cell.

Choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and II both are incorrect.

9. **Which of the following organelle (s) is/are not include in endomembrane system?**

(1) Endoplasmic Reticulum

(2) Golgi body

(3) Lysosome and vacuoles

(4) Mitochondria, chloroplast and ribosome

10. **The incorrect statement about lysosome is:**

(1) single membrane bounded.

(2) presence of hydrolytic enzymes.

(3) acidic pH due to outgoing of protons in cytoplasm.

(4) formed by packaging in the Golgi body.

11. **In plants, the ____A___ facilitates the transport of a number of ions and other materials against concentration gradient into the ___B__.**

(1) A - Tonoplast B - Vacuole

(2) A - Outer membrane B - Mitochondria

(3) A - Outer membrane B - Chloroplast

(4) A - Tonoplast B - Mitochondria

12. **Which of the following structure contains respiratory enzymes?**

(1) Cell wall

(2) Mesosome

(3) Chromatophore

(4) Plasmid

13. **Given below are two statements :**

Statement I : Ribosomes are non membrane-bound organelles found in all cells – both eukaryotic and prokaryotic.

Statement II : Within the cells, ribosomes are found not only in the cytoplasm but also within the two organelles- chloroplast and mitochondria and on rough ER.

(1) Both statements I and II are correct.

(2) Both statements I and II are incorrect.

(3) Only statement I is correct but statement II is incorrect.

(4) Only statement I is incorrect but statement II is correct.

14. **Which of the following statement(s) of a bacterial cell is/are correct?**

(i) Mesosomes are formed by extensions of plasma membrane into the cell.

(ii) The pili are elongated tubular structures made up of a proteins.

(iii) Flagellum is composed of filament, hook and basal body.

(iv) Ribosomes are about 30 nm by 50 nm in size

(1) (i), (ii) and (iii)

(2) (iii) and (iv)

(3) (ii) and (iv)

(4) None of the above

15. **Match column I with column II**

Column I	Column II
A. Double membrane	**i. Vacuole**
B. Single membrane	**ii. Mitochondria**
C. Non-membranous	**iii. Mesosomes**
D. Extensions of plasma membrane	**iv. Ribosome**

(1) A-ii, B-i, C-iv, D-iii

(2) A-iii, B-iv, C-ii, D-i

(3) A-iv, B-iii, C-i, D-ii

(4) A-iii, B-i, C-iv, D-ii

16. Given below are two statements;

Assertion (A): Inclusion bodies are not bound by any membranous system and do not lie free in the cytoplasm.

Reason (R): Reserve material in prokaryotic cells is stored in the cytoplasm in the form of inclusion bodies.

Choose the most appropriate answer from the options given below :

(1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) and Reason (R) both are false.

17. Which statement provides correct information regarding prokaryotic and eukaryotic flagella:

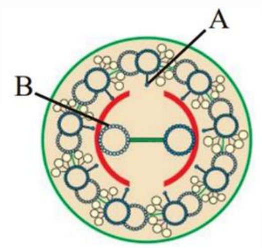
(1) The prokaryotic cell also possess flagella but these are structurally different from that of eukaryotic flagella.

(2) Prokaryotic and eukaryotic both flagella are structurally similar.

(3) In case of both microtubule arrangement is 9 + 2.

(4) In case of both flagella is made up of microtubule and flagellin protein.

18. The figure below shows the section of a cilia/flagella showing different parts labelled A and B select the correct labelled name ?



(1) A – Radial spoke B – Central microtubule

(2) A – Interdoublet Bridge B – Peripheral microtubule

(3) A – Centeral sheath B – Plasma membrane

(4) A – Interdoublet Bridge B – Radial spoke

19. Given below are two statements;

Assertion (A) : Chromatin are nucleoprotein fibres.

Reason (B) : Chromatin contains DNA and some basic proteins called histones, some non-histone proteins and also RNA.

Choose the most appropriate answer

(1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true but Reason (R) is not correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) and Reason (R) both are false.

20.What is the primary role of nuclear pores in the nuclear envelope?

(1) Synthesis of DNA

(2) Facilitate the movement of molecules between the nucleus and cytoplasm

(3) Store cellular waste

(4) Generate ATP

21.Mark the incorrect one.

(1) Robert Brown described nucleus in 1831.

(2) Outer nuclear membrane is continuous with endoplasmic reticulum.

(3) outer nuclear membrane also bears ribosomes on it.

(4) The 1 nm – 5 nm broad perinuclear space separates both membranes of nucleus.

22.Match the Column-I and Column-II

Column - I	Column - II
(A) Metacentric	**(i) At the tip**
(B) Sub-Metacentric	**(ii) At the middle**
(C) Acrocentric	**(iii) Almost near the tip**
(D) Telocentric	**(iv) Slightly away from the middle**

(1) A - ii, B - iv, C - i, D - iii

(2) A - ii, B - iv, C - iii, D - i

(3) A - iv, B - ii, C - iii, D - i

(4) A - i, B - iii, C - ii, D – iv

23. Given below are two statements;

Assertion (A): Nucleolus is the site for active ribosomal RNA synthesis.

Reason (R): Larger and more numerous nucleoli are present in cells actively carrying out lipid synthesis.

Choose the most appropriate answer from the options given below:

(1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) and Reason (R) both are false.

24.Which of the following cytoskeleton is a structural component of centrioles?

(1) Microtubules

(2) Microfilaments

(3) Intermediate filaments

(4) All of the above

25.The shorter and longer arms of a submetacentric chromosome are referred to as:

(1) s-arm and l-arm respectively.

(2) p-arm and q-arm respectively.

(3) q-arm and p-arm respectively.

(4) m-arm and n-arm respectively.

26.Given below are the two statements:

Statement I : The core of cilium or flagellum is called axoneme.

Statement II : Central tubules are connected by bridge and enclosed by central sheath.

Choose the most appropriate answer

(1) Statement-I is correct but Statement-II incorrect.

(2) Statement-I is incorrect but Statement II is correct.

(3) Statement-I and Statement-II both are incorrect.

(4) Statement-I and Statement-II both are correct.

27. The chemical studies on cell membrane that helped to deduce its possible structure was mostly done on:

(1) Human red blood cell

(2) WBC

(3) Platelets

(4) None

28. A large central vacuole is present in:

(1) prokaryotic cell

(2) animal cell

(3) plant cell

(4) both plant and animal cell.

29.Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R)

Assertion (A) : Fluid mosaic model is most widely accepted model of plasma membrane.

Reason (R) : It states that protein icebergs are present in mosaic pattern in the sea of glycolipid.

In the light of the above statements, Choose the most appropriate answer from the options given below :

(1) If (A) & (R) both are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct.

(4) If (A) & (R) both are correct and (R) is the correct explanation of (A).

30. Which option is incorrect about chloroplast?

(1) The length is $5\mu m - 10\mu m$ while width is $2\mu m - 4\mu m$.

(2) Flat membranous tubules called stroma lamellae connect thylakoids of the different grana.

(3) Chlorophyll pigments are present in stroma

(4) More than one options are incorrect.

31. Select the correct option .

(1) The shape of cell may not vary with function they perform.

(2) Cell wall is unique feature of animal cell.

(3) 'Omnis cellula e cellula' was proposed in 1855.

(4) Tracheid cell is typical animal cell.

32.Which of the following features is common to prokaryotes and many eukaryotes?

(1) Chromatin material present.

(2) Cell wall present.

(3) Nuclear membrane present.

(4) Membrane bound sub-cellular organelles present.

33. Given below are two statements :

Assertion (A) : Cell is the fundamental structural & functional unit of life.

Reason (B) : Unicellular organisms are capable of independent existence.

(1) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) and Reason (R) both are false.

34. Given below are two statements :

Statement I : All living organisms are composed of cells and products of cells.

Statement II : All cells arise from pre-existing cells.

Choose the most appropriate answer from the options given below :

(1) Statement-I and Statement-II both are correct.

(2) Statement-I is incorrect but Statement II is correct.

(3) Statement-I and Statement-II both are incorrect.

(4) Statement-I is correct but Statement-II is incorrect.

35.Match between column I and column II

Column-I	Column-II
(A) Typical Bacteria	**(i) 10 - 20 µm**
(B) PPLO	**(ii) 0.02 - 0.2 µm**
(C) Viruses	**(iii) 0.1 µm**
(D) A Typical eukaryotic cell	**(iv) 1 - 2 µm**

Choose the correct answer

(1) A-iv B-iii C-ii D-i

(2) A-iv B-ii C-i D-iii

(3) A-i B-ii C-iii D-iv

(4) A-iii B-iv C-ii D-i

36. Identify the correct set of statements out of followings :-

(a) Maximum arm ratio exhibited by acrocentric chromosomes

(b) Human cells (2n) has two genomes and each genome consists fourty six chromosomes

(c) Nucleolar organizer regions are not present on each human chromosomes

(d) Secondary constriction is present at the place of centromere

(e) Each centromere posses two protein discs called kinetochores

Choose the correct answer from the given options below.

(1) b, c, d & e

(2) a, b, c & d

(3) a, c & e

(4) a, b & c

37.Which option provide correct information regarding the nucleus.

(1) Normally, there is only one nucleus per cell, variation in the number of nuclei are also observed.

(2) Normally, there is multiple nucleus per cell.

(3) Chlamydomonas contains only one nucleus.

(4) 1 and 3 option is correct.

38. Identify the wrong statement.

(1) Cilia work like oars

(2) Both cilia & flagella are hair like out growth of cell membrane in eukaryotes.

(3) Both prokaryotic and eukaryotic flagella are structurally and functionally similar

(4) All of the above

39. Find out the odd one about centrioles

(1) The centrioles give rise to spindle apparatus during cell division in animal cells.

(2) Each of the peripheral fibril is a doublet.

(3) Centrioles are surrounded by amorphous pericentriolar materials.

(4) Centrioles have an organization like the cartwheel.

40. Identify the incorrect statement.

(1) The nuclear matrix or the nucleoplasm contains nucleolus and chromatin.

(2) The nucleoli are spherical structures present in the nucleoplasm.

(3) Material of the nucleus stained by the basic dyes was given the name chromatin by Flemming.

(4) The interphase nucleus has highly condensed and elaborate nucleoprotein fibres called chromosomes.

41. Given below are two statements;

Statement I : Mitochondria are the sites of anaerobic respiration.

Statement II : The mitochondrial matrix also possess single circular DNA molecule, a few RNA molecules, ribosomes (80 S) and the components required for the synthesis of proteins.

(1) Both statement I and II are correct.

(2) Statement I is correct but, Statement II is incorrect.

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and II are incorrect.

42. Given below are two statements;

Assertion : Chromoplast is coloured plastid in corolla and fruit.

Reason : It has water soluble chlorophyll and carotenoid pigments.

Choose the most appropriate answer from the options given below:

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion

(2) Assertion is correct but Reason is not correct.

(3) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

(4) Assertion is not correct but Reason is correct.

43. Which of the following features is present in mitochondria?

(1) Linear DNA, 70S ribosome

(2) Circular DNA, glycogen synthesis

(3) Single circular DNA,70S ribosome

(4) Many circular DNA, single-stranded DNA, 70S ribosome

44. Which similarity is found between chloroplast and mitochondria ?

(1) Both are semi-autonomous contain double stranded circular DNA and double membrane bound.

(2) Both are autonomous organelle.

(3) Both are single membrane bounded.

(4) Both contains 80S type of ribosome.

45. Given below are two statements:

Statement-I : Membrane of lysosome is known as tonoplast.

Statement-II : Contractile vacuole helps in osmoregulation.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect & Statement II is correct.

(4) Both Statement-I and II both are incorrect.

46. Which is mismatched pair?

(1) Capsule - Thick and tough glycocalyx

(2) Slime layer - Loose glycocalyx

(3) Pili - Motility organ

(4) Bacterial cells - Motile or nonmotile

47. All of the following statements are correct about plasmids except:

(1) they are extrachromosomal DNA.

(2) they are smaller, circular, double stranded naked DNA that confer certain unique phenotypic characters to some bacteria like resistance to antibiotics.

(3) they are used in genetic engineering.

(4) it helps in the replication of nucleoid.

48. Given below are two statements;

Statement I : Animal cells have centrioles which are absent in almost all plant cells.

Statement II : Plant cells possess cell walls, plastids which are absent in animal cells.

Choose the correct option

(1) Both statements I and II are correct.

(2) Both statements I and II are incorrect.

(3) Only statement I is correct but statement II is incorrect.

(4) Only statement I is incorrect but statement II is correct.

49. Na+/K+pump is involved in which type of transport?

(1) Downhill

(2) Passive transport

(3) Active transport

(4) Along concentration gradient

50. In a prokaryotic cell –

A. Enveloped genetic material is present.

B. Ribosomes are absent.

C. An organized nucleus is absent.

The correct option is :

(1) Only A

(2) Only B

(3) Only C

(4) All of these

Answer Key

1. (2)
2. (4)
3. (1)
4. (1)
5. (4)
6. (3)
7. (2)
8. (2)
9. (4)
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47. (4)
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49. (3)
50. (3)